25901395

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECMAT200 - DOCUMENT PREPARATION USING LATEX

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 15 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type Questions

Answer any **5** questions

Each question carries **2** marks

1. What command is used to start a new item in a list? [U] / [CO1]

2. What type of file is used as input for $LaTeX$? [U] / [CO1]

3. Write the LaTeX command to type the symbols \, { and } [S] / [CO2]

4. Which pair of commands do you use in the document body to activate a user-defined environment? [U] / [CO3]

5. Which two environments allow LaTeX to move content to the top or bottom of a page to avoid gaps? [U] / [CO2]

6. What is the purpose of the overlay environment in slides? [S] / [CO5]

7. Write the output of the folowing $LaTeX$ code [C] / [CO4]

```
\begin{thebibliography}{99}
\bibitem{a1}   Ponnusamy, S., Herb Silverman. \textbf{Complex variables with applications. Springer Science & Business Media}, 2007.
\end{thebibliography}
```

8. What is the function of the \nocite command? [S] / [CO4]

[2x5 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

9. Write a short note on math styles in LaTeX? What decalartions are used to force each math style? [A] / [CO2]

10. With proper examples write the purpose of the following LaTeX commands [S] / [CO3]
    1. `\upshape`
    2. `\itshape`
    3. `\slshape`

11. Describe how LaTeX places floating objects in a document. [U] / [CO2]

12. Write the output of the following LaTeX code [U] / [CO3]

```
\newtheorem{thm}{Theorem}
\begin{thm}[Euclid]
The sum of the angles of a triangle is 180 degree
\end{thm}
```

13. Explain with an example, how can we give `\label` command in `eqnarray environment`, [A] / [CO3]

14. Explain the use of the folowwing commands in book class [S] / [CO2]
    1. \openany
    2. \openright
    3. \backmatter

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

15. Give the LaTeX commands for following mathematical formuals [S] / [CO2]

    (a) $\sqrt[10]{x^3}$

    (b) $\frac{y+z/2}{y^2+1}$

    (c) $\sum_{i=1}^{n} x_i = \int_0^{\pi} f$

16. (a) Write LaTeX file to produce the following output (without outer frame) [A] / [CO2]

Thus x, y and z satisfy the equations

$$x + y - z = 1$$
$$x - y + z = 1$$

and by hypothesis

$$x + y + z = 1 \qquad (1)$$

(b) Write the output of the following $LaTeX$ code

```
\begin{eqnarray}
(a+b+c+d+e+f)^2 & = a^2+b^2+c^2+d^2+e^2+f^2
& +2ab+2ac+2ad+2ae+2af\\
& +2bc+2bd+2be+2bf\\
& +2cd+2ce+2cf\\
&+2de+2df\\
& +2ef
\end{eqnarray}
```

17. **(a)** Explain how `\newcommand` helps in $LaTeX$. [A] /
(b) Write the $LaTeX$ code using `\newcommand` to create a shortcut `\smat` for the course name "Skill Enhancement Course in Mathematics". [CO3]
(c) Write the output of the following $LaTeX$ code

```
\newcommand{\vect}[2]{(#1_1,#1_2,\dotsc,#1_#2)}
$\vect{x}{n},\vect{y}{m}, \vect{z}{k}$
```

18. Write LaTeX code to produce an equation with a label and refer to it later in the text. Show the expected output. [S] / [CO3]

[10x2 = 20]